# Lab 05

## Predicting a numeric outcome

Teaching Week 05 | Individual work. Submit in R or Python.

**Purpose**

This lab applies this week's evaluation metrics to a numeric prediction task. You will define a target, build two simple baseline predictions, score them with several metrics, and interpret the errors in the target's original units.

## Learning outcomes

After completing this lab, you should be able to

- distinguish prediction from inference;
- construct simple baseline predictions for a numeric target;
- compare prediction rules using several error metrics, including a relative metric; and
- diagnose where prediction errors are largest.

## Dataset choice

Choose one of the verified open datasets below. Each option supports the assessment when your analytical table satisfies the data contract in the next section. The expected variables describe the published record and may differ in detail from the extract you retrieve; reconcile any difference in your report.

**Option A | Halifax County Water Quality Data**

**Publisher.** Nova Scotia Fisheries and Aquaculture
**Official record.** https://data.novascotia.ca/d/x9dy-aai9
**Record identifier.** `x9dy-aai9`
**Licence and format.** Nova Scotia Open Government Licence; CSV
**Expected variables.** waterbody; station; lease; timestamp; latitude; longitude; temperature; salinity; dissolved oxygen; sensor depths; quality flags
**Use in this lab.** predict a selected environmental measurement
**Why it fits.** Repeated numeric sensor measurements support station and season baselines and error metrics.

**Option B | Fuel Consumption Ratings**

**Publisher.** Natural Resources Canada
**Official record.** https://open.canada.ca/data/en/dataset/98f1a129-f628-4ce4-b24d-6f16bf24dd64
**Record identifier.** `98f1a129-f628-4ce4-b24d-6f16bf24dd64`
**Licence and format.** Open Government Licence – Canada; CSV
**Expected variables.** model year; make; vehicle class; engine size; cylinders; fuel type; fuel consumption; CO2 emissions
**Use in this lab.** predict fuel consumption or carbon dioxide emissions
**Why it fits.** Canadian energy-efficiency and emissions records support vehicle-class baselines and errors in meaningful units.

**Option C | Energy consumption of home appliances - EnerGuide label**

**Publisher.** Natural Resources Canada
**Official record.** https://open.canada.ca/data/en/dataset/fbfdf946-8dd1-4830-a5c9-f8a72d8fabda
**Record identifier.** `fbfdf946-8dd1-4830-a5c9-f8a72d8fabda`
**Licence and format.** Open Government Licence – Canada; ZIP archives containing CSV files
**Expected variables.** appliance type; brand; model; capacity; energy use
**Use in this lab.** predict energy consumption from product attributes
**Why it fits.** Numeric and categorical predictors support baselines and comparative metrics.

## Data contract

Choose one listed dataset and define one numeric prediction target.

- Retain at least 200 complete observations after documented cleaning.
- Include one numeric target and one category, station, season, or time column that can define a group baseline.
- Compare a constant baseline (the mean or median of the target) with a group baseline (the mean of the target within each group).
- Score both baselines on the same evaluation records, chosen by one stated rule such as the most recent year, and compute both from the remaining records.
- Report at least two appropriate error metrics, including one relative metric that names its baseline, and interpret their units.
- Reuse prior summaries, cleaning, graphics, and reproducible reporting.
- Document the official record URL, publisher, licence, retrieval steps, and access date.

**What the contract does not settle**

The contract constrains the table. It does not choose the target, the group column, or the metrics.

## Selection checklist

Resolve every unchecked item before you begin the analysis.

- ☐ I chose one option from the dataset catalogue above.
- ☐ I recorded the official URL, publisher, licence, access date, and retrieval steps.
- ☐ I defined one numeric target and one group column for a group baseline.
- ☐ My table satisfies every rule in the data contract.
- ☐ I stated the rule that selects the evaluation records.
- ☐ I selected a constant baseline, a group baseline, and two interpretable error metrics.
- ☐ I checked the target for zeros before choosing a percentage metric.
- ☐ I will cite the dataset in the report.

## Cumulative method

This lab carries forward the earlier data audit, cleaning pipeline, statistical graphics, and reproducible report structure. Your report must show how those earlier decisions support the current analysis rather than repeating them as disconnected exercises.

## Work program

### 1. Define the prediction problem

State the target, the prediction unit, the group column, the use case, and whether the goal is prediction or inference.

### 2. Build the baselines

Select the evaluation records by one stated rule. Compute the constant baseline and the group baseline from the remaining records, and say which mean or median each one uses.

### 3. Evaluate errors

Report at least two appropriate metrics for both baselines, including one relative metric, a predicted-versus-observed plot, and error summaries for one meaningful subgroup or range.

### 4. Compare the metrics

Explain which metric suits the use case and why. Where two metrics rank the baselines differently, say which errors cause the disagreement.

### 5. State the claim

Explain what the errors would mean in use, where the baselines fail, and why predictive association is not a causal explanation.

## Reproducible workflow

You may complete and submit the lab in either R or Python. Both paths are marked against the same evidence and rubric, and both must run end to end from one documented entry point in your portfolio repository.

Write your own analysis using the methods taught in the course. Document how to run it from the retained source data and produce every result cited in your report. Choose and justify the variables, preparation steps, and analytical methods required by your question and this lab's data contract. Record the software environment and any randomness that affects reproducibility.

Downloaded data and generated results are not course source, so they never go into the course repository. Commit the report, its source, and the generated tables and figures it cites to your private work repository. Keep that evidence private until the deadline, and publish it in your public GitHub portfolio repository only when the instructor opens sharing. Link rather than redistribute source data unless its licence and size permit it.

## Submission package

Keep the lab private before its submission deadline, including cloud notebooks and answer-revealing links. A branch in a public repository is not private. Give the instructor access to your private work repository. First commit the completed report, source, and supporting evidence. Record that commit's full SHA and commit URL in `labs/lab-05/README.md`, then commit this README update separately. A commit cannot contain its own SHA. Push both commits to the default branch before 16:05 Halifax time on October 15, 2026, seven calendar days after the integrated lab. The deadline holds even when no class meets that day. The marker reads that README on the default branch and assesses the snapshot it names. The immutable commit or release URL must resolve to that exact version of `labs/lab-05/`. A branch URL or movable tag alone is not a fixed submission. The package includes

- a concise PDF report that answers the analytical question, together with the completed selection checklist;
- the complete R or Python source plus the generated tables, figures, and any summary statistics referenced in the report;
- a README carrying the submitted commit SHA and its commit URL, the official data URL, retrieval instructions, exact execution command, and environment; and
- no redistributed source data unless its licence and size permit it.

After the deadline and when the instructor opens sharing, publish the same submitted commit and its history in your public GitHub portfolio repository and present that version at the next class meeting, if one follows. Preserve the submitted SHA when merging. Commit regularly as you work. Do not amend, rebase, squash, delete, replace, force-push, or backdate recorded assessment history. Keep submitted commits reachable and make corrections in new commits. The pushed README record and the instructor-retained snapshot provide the independent assessment reference; Git author dates alone are not proof of timely submission.

## Assessment standard

The course uses one rubric across all integrated labs. The best eight of the nine weekly lab results contribute 15% of the course grade.

**Table 1.** Integrated lab marking criteria.

| Criterion | Weight (%) |
|---|---|
| Dataset provenance and suitability | 15 |
| Reuse of earlier methods | 20 |
| Current-week analytical method | 35 |
| Interpretation and limitations | 20 |
| Reproducibility and communication | 10 |
| Total | 100 |

Academic integrity, private GitHub exceptions, private grading, and publication limits follow the course syllabus. Generative AI output is prohibited on integrated labs, so generated code, prose, analysis, or figures cannot be submitted for lab credit. Cite the dataset and every non-trivial external source; the submitted analysis must be your own work.